

CHAPTER 4 ACCESS CONTROL

- 4.1** Early Intel processors (e.g., the 8086) did not provide hardware support for dual-mode operation (i.e., support for a separate user mode and kernel mode). As a result, most of the systems implemented on these processors did not support multi-user operation. List and explain one potential problem associated with supporting multi-user operation without hardware support for dual-mode operation
- 4.2** State True/False and also give justification for each statement.
- a.** Access control matrices can represent anything that is represented by access control lists.
 - b.** Consider data that is stored over time in a mandatory access control based system. The contents of files containing highly classified ("top secret") information are necessarily more trustworthy than material stored in files marked unclassified
- 4.3**
- a.** Briefly define what a "Role-Based Access Control (RBAC)" system is, and its key components.
 - b.** State how RBAC differs from a Discretionary Access Control (DAC) system.
 - c.** What reasons might lead an organization to implement an RBAC system rather than the older DAC system?